

Near-linear, but never linear: a universal ceiling for consecutive log-concavity failures

Research draft prepared for Jan Mikulík with ChatGPT

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Abstract

For every fixed maximum degree Δ , consecutive runs of strict log-concavity failures in forest independence polynomials have sublinear length. We prove the effective upper bound $M_\Delta(N) < 6N/\log_2 \log_{\Delta-1} N$ for $N \geq (\Delta-1)^{2^{14}}$, where $\Delta \geq 3$ and $M_\Delta(N)$ maximizes over forests with at most N vertices. The argument combines an exact positive-coefficient identity converting a convex probability window into a variance lower bound, entropy control of the appropriate activity, and covariance decay along tree paths. At any fixed proportional distance below the independence number, the run length instead satisfies an explicit polynomial saving. Together with the companion degree-65 construction, this gives exponent one but zero asymptotic density.

1 The universal statement

Let $i_k(F)$ count independent sets of size k in a finite forest F , and let $\alpha(F)$ be its independence number. A strict failure occurs at k if $i_k(F)^2 < i_{k-1}(F)i_{k+1}(F)$. A run is a consecutive interval of such indices, and its length is the number of failure indices, not the number of coefficients in the enlarged window. For integer $\Delta \geq 3$, define

$$M_\Delta^{\text{for}}(N) = \max_{\substack{|V(F)| \leq N \\ \Delta(F) \leq \Delta}} \{\text{length of a strict failure run in } F\}.$$

The maximum is zero if no failure occurs. Restricting the maximum to trees gives $M_\Delta^{\text{tree}}(N) \leq M_\Delta^{\text{for}}(N)$.

Theorem 1. *Let $b = \Delta - 1 \geq 2$. For each integer $m \geq 1$, set*

$$\Lambda_m = 4^m, \quad K_m = \lceil \log_2(12m^2) \rceil, \quad R_m = (\Lambda_m + 1)K_m. \quad (1)$$

Every n -vertex forest of maximum degree at most Δ with

$$n \geq n_{\Delta,m} := 36m^2 b^{R_m}$$

has no failure run of length L satisfying $L + 1 \geq n/m$. Consequently, for every

$$N \geq N_{\Delta,m} := 36m^3 b^{R_m}, \quad M_\Delta^{\text{for}}(N) < N/m. \quad (2)$$

Corollary 2. *For each $\Delta \geq 3$,*

$$\boxed{M_\Delta^{\text{for}}(N) < \frac{6N}{\log_2 \log_{\Delta-1} N} \quad \text{for } N \geq (\Delta-1)^{2^{14}}.} \quad (3)$$

In particular $M_\Delta^{\text{for}}(N) = o(N)$. For $\Delta = 65$, the displayed threshold is the explicit integer $N \geq 2^{98304}$.

The large threshold is deliberately conservative. It is a closed formula, not an unevaluated “sufficiently large” constant. Neither this upper bound nor its constant is claimed to be asymptotically sharp. All probabilistic estimates used below hold at finite order and finite positive activity; there is no local limit theorem in this proof.

2 An exact variance certificate for a convex window

Lemma 3. *Let X be integer-valued. Suppose its probabilities p_j are convex on the integer interval $[a, b]$, meaning $p_{j-1} - 2p_j + p_{j+1} \geq 0$ for $a < j < b$. Put $d = b - a \geq 2$ and $c = (a + b)/2$. Then*

$$\mathbb{E}(X - c)^2 \geq \frac{d(d+2)}{12}. \quad (4)$$

Thus if $\mathbb{E}X = c$, the same bound holds for $\text{Var } X$.

Proof. For an arbitrary sequence $u_0, \dots, u_d$, the exact identity

$$\begin{aligned} & \sum_{j=0}^d \left\{ (j - d/2)^2 - \frac{d(d+2)}{12} \right\} u_j \\ &= \sum_{k=1}^{d-1} \underbrace{\frac{k(k+1)(d-k)(d-k+1)}{12}}_{C_{d,k} > 0} (u_{k-1} - 2u_k + u_{k+1}) \end{aligned} \quad (5)$$

follows by collecting coefficients. Specifically, with $C_{d,0} = C_{d,d} = 0$,

$$C_{d,k-1} - 2C_{d,k} + C_{d,k+1} = (k - d/2)^2 - \frac{d(d+2)}{12} \quad (1 \leq k \leq d-1),$$

and the two endpoint coefficients agree as well. The companion program verifies these as polynomial identities in two indeterminates, rather than checking finitely many window lengths.

Apply (5) with $u_j = p_{a+j}$. Its right side is nonnegative. For an integer j outside $[a, b]$, $|j - c| \geq d/2 + 1$ and

$$(j - c)^2 - \frac{d(d+2)}{12} \geq \frac{(d+2)(d+3)}{6} > 0.$$

Adding the outside probabilities proves (4). $\square$

From a failure run to the lemma. Suppose $i_k^2 < i_{k-1}i_{k+1}$ for $k = u, \dots, v$, a run of length $L = v - u + 1$. Set

$$a = u - 1, \quad b = v + 1, \quad d = L + 1, \quad c = (u + v)/2. \quad (6)$$

For any $\lambda > 0$, the tilted probabilities

$$p_k = \mathbb{P}_\lambda(X = k) = \frac{i_k \lambda^k}{Z_F(\lambda)}, \quad Z_F(\lambda) = \sum_j i_j \lambda^j,$$

obey the same strict log-convexity inequalities on this window. Since $p_{k-1} + p_{k+1} \geq 2\sqrt{p_{k-1}p_{k+1}} > 2p_k$, they are strictly convex there. The function $\mathbb{E}_\lambda X$ increases continuously from 0 to $\alpha(F)$ as λ increases from 0 to infinity, because

$$\frac{d}{d \log \lambda} \mathbb{E}_\lambda X = \text{Var}_\lambda X > 0.$$

There is therefore a unique λ for which $\mathbb{E}_\lambda X = c$. At that activity, Lemma 3 yields

$$\text{Var}_\lambda X \geq \frac{(L+1)(L+3)}{12} > \frac{(L+1)^2}{12}. \quad (7)$$

This argument needs no assumptions on the coefficients outside the run.

3 Entropy controls the activity

For the Gibbs distribution on independent sets, its Shannon entropy in nats is

$$H = \log Z_F(\lambda) - (\mathbb{E}_\lambda X) \log \lambda.$$

Since $Z_F(\lambda) \geq \lambda^{\alpha(F)}$ and the number of independent sets is at most 2^n ,

$$(\alpha(F) - \mathbb{E}_\lambda X) \log \lambda \leq H \leq n \log 2. \quad (8)$$

This inequality holds also when $\lambda < 1$. For the centered activity associated with (6), $\alpha(F) - c \geq d/2$, since $b \leq \alpha(F)$. Hence

$$\lambda \leq 2^{2n/d}. \quad (9)$$

In particular $d \geq n/m$ implies $\lambda \leq 4^m = \Lambda_m$.

4 Covariance decay on every finite forest

Lemma 4. *Let F have n vertices and maximum degree at most $\Delta \geq 3$. Put $b = \Delta - 1$ and $q = \lambda/(1 + \lambda)$. For every integer $R \geq 0$,*

$$\text{Var}_\lambda X \leq \frac{3nb^R + n^2q^R}{4}. \quad (10)$$

Proof. Write $S_v = \mathbf{1}_{\{v \in I\}}$. Root a tree component at u . For a directed parent-child edge $w \rightarrow z$, let q_z be the occupation probability of z conditional on $S_w = 0$. The separated subtree has root odds at most λ , so $0 < q_z \leq q$. The exact transition is

$$\mathbb{E}(S_z \mid S_w) = q_z(1 - S_w).$$

Removing an edge separates the component, giving the Markov property along any path. If $u = v_0, v_1, \dots, v_t = v$, composition of the affine conditional expectations therefore gives

$$\text{Cov}(S_u, S_v) = (-1)^t \text{Var}(S_u) \prod_{j=1}^t q_{v_j}, \quad |\text{Cov}(S_u, S_v)| \leq \frac{1}{4} q^t. \quad (11)$$

Distinct components are independent, so their covariances are zero.

The ball of radius R about a vertex has at most

$$1 + \Delta \sum_{j=0}^{R-1} (\Delta - 1)^j \leq 3b^R$$

vertices (the sum is empty when $R = 0$). For $R \geq 0$, the difference between $3b^R$ and the geometric expression, multiplied by $b - 1$, is $(2b - 4)b^R + 2 \geq 0$. Sum (11) over ordered pairs. There are at most $3nb^R$ near pairs; all other pairs have covariance modulus at most $q^R/4$, or zero. Bounding the variance by the sum of covariance moduli proves (10). $\square$

5 Proof of the finite threshold and its explicit envelope

Suppose $d = L + 1 \geq n/m$ and choose the centered activity. By (9), $q \leq \Lambda_m/(1 + \Lambda_m)$. The integer binomial inequality

$$(1 + 1/\Lambda_m)^{\Lambda_m+1} \geq 1 + (\Lambda_m + 1)/\Lambda_m > 2$$

implies, with the exact radius in (1),

$$q^{R_m} < 2^{-K_m} \leq \frac{1}{12m^2}. \quad (12)$$

If $n \geq 36m^2b^{R_m}$, Lemma 4 gives

$$\text{Var}_\lambda X \leq \frac{n^2}{48m^2} + \frac{n^2}{48m^2} = \frac{n^2}{24m^2} < \frac{d^2}{12},$$

contradicting (7). This proves the exact-order assertion in Theorem 1. For $N \geq mn_{\Delta,m}$, a forest with $n < n_{\Delta,m}$ has every run shorter than N/m simply because $L \leq n$. Forests with $n \geq n_{\Delta,m}$ satisfy $L < n/m \leq N/m$. This proves the at-most-order assertion (2). $\square$

Deriving the closed $N/\log \log N$ bound. For every integer $m \geq 1$,

$$K_m \leq 4m, \quad m \leq 2^m, \quad R_m \leq 8m4^m \leq 2^{3m+3}.$$

The first inequality follows from $12m^2 \leq 16m^2$: it holds at $m = 1$, and $16m^2 - (m+1)^2 > 0$ for $m \geq 1$ preserves it. Also $36m^3 \leq 2^{6+3m}$ and $6+3m \leq 2^{3m+3}$. Consequently

$$N_{\Delta,m} = 36m^3b^{R_m} \leq b^{R_m+6+3m} \leq b^{2^{3m+4}}. \quad (13)$$

These inequalities are proved by elementary induction, and their algebraic witnesses are included in the verifier.

For $N \geq b^{2^{14}}$, let $A = \log_2 \log_b N \geq 14$ and $m = \lfloor (A-4)/3 \rfloor$. Then $3m+4 \leq A$ and

$$m \geq \frac{A-7}{3} \geq \frac{A}{6}.$$

Equation (13) permits applying Theorem 1, yielding $M_\Delta^{\text{for}}(N) < N/m \leq 6N/A$. This proves Corollary 2. $\square$

6 A polynomial barrier away from the final coefficient

The same argument also forces runs of exponent tending to one to move towards the independence number.

Theorem 5. Fix $\Delta \geq 3$, put $b = \Delta - 1$, and let $0 < \delta < 1$. Define

$$\varepsilon_{\Delta,\delta} = \frac{1}{8(1+2^{2/\delta})\log b} > 0.$$

If a failure run of length L in an n -vertex forest has midpoint $c \leq (1-\delta)\alpha(F)$ and $n \geq b^4$, then

$$L+1 < \sqrt{12}n^{1-\varepsilon_{\Delta,\delta}}. \quad (14)$$

Proof. A forest is bipartite, so $\alpha(F) \geq n/2$. At the centered activity, (8) gives $\lambda \leq \Lambda := 2^{2/\delta}$. Take $R = \lfloor \frac{1}{2} \log_b n \rfloor$; since $n \geq b^4$,

$$b^R \leq \sqrt{n}, \quad R \geq \frac{\log n}{4 \log b}.$$

The elementary bound $\log(1+1/\Lambda) \geq 1/(1+\Lambda)$ yields

$$q^R \leq n^{-\eta}, \quad \eta = \frac{1}{4(1+\Lambda)\log b} = 2\varepsilon_{\Delta,\delta}.$$

Here $0 < \eta < 1/2$. Thus (10) gives

$$\text{Var}_\lambda X \leq \frac{3}{4}n^{3/2} + \frac{1}{4}n^{2-\eta} \leq n^{2-\eta}.$$

Combine this with (7). $\square$

In particular, for any sequence of bounded-degree forests with $n \rightarrow \infty$ and failure runs of length $L = n^{1-o(1)}$, their midpoints satisfy $c/\alpha(F) \rightarrow 1$. Otherwise a fixed $\delta > 0$ and an infinite subsequence would contradict (14). By Corollary 2, $L = o(n)$; since $\alpha(F) \geq n/2$, both endpoints of each such run also have ratio tending to one with $\alpha(F)$. Thus near-linear exponents cannot persist in a fixed proportional bulk interval, under any bounded-degree construction.

7 Consequence for the degree-65 construction

The companion manuscript [1] proves $M_{65}^{\text{tree}}(N) = N^{1-o(1)}$ using even-path seeds and 64-fold branching. Combining that lower bound with the independent upper bound proved here gives

$$\lim_{N \rightarrow \infty} \frac{\log M_{65}^{\text{tree}}(N)}{\log N} = 1, \quad \lim_{N \rightarrow \infty} \frac{M_{65}^{\text{tree}}(N)}{N} = 0. \quad (15)$$

The first limit uses the companion construction. The second limit, and the explicit upper bounds in this paper, do not depend on it. The linear-run target is therefore impossible at any fixed maximum degree. Equation (15) does not determine the exact subpolynomial factor between the upper and lower bounds.

8 Verification scope and literature

The proof above is finite and self-contained. The file `verify_sublinear_ceiling.py` uses only the Python standard library and exact rational arithmetic. Running

```
python verify_sublinear_ceiling.py
```

writes `sublinear_ceiling_certificate.json`. The executed run passed 18 universal algebra/constant checks and an independent exact enumeration audit of 1364 ordered covariance pairs, with 5245 checks in total. The algebra checks include the two-variable polynomial identity (5) and induction witnesses for (13). They are not numerical samples of the indeterminates. The enumeration audit uses five small forests at four rational activities and is explicitly a finite sanity check. The probability, entropy, positivity and induction arguments establishing the universal quantifiers are the written proofs above; no proof-assistant formalization is claimed.

For orientation, Yu’s September 2026 archive [2] supplies unbounded consecutive runs and an $\Omega(N^{1/3})$ lower bound. Fang, Lu, Nevo, Yao and Zheng [3] establish eventual unimodality of forest independence sequences via central-interval log-concavity. Their proof uses a fixed activity range $[1/4, 12]$. The present upper bound concerns the length of an arbitrary run and allows the activity to depend on its relative length; it does not extend their local-limit theorem to all activities. The primitives used here—Gibbs tilting, entropy and tree conditional independence—are standard. This literature comparison is not a claim of exhaustive publication priority. This text is a research draft, not an independently refereed or formally verified result.

References

- [1] *Near-linear runs of strict log-concavity failures at one fixed maximum degree*. Companion research draft prepared for Jan Mikulík with ChatGPT, 28 September 2026. Source file: `fixed_degree_nearlinear.tex`.
- [2] X. Yu, *Unbounded runs of non-log-concavity in independence polynomials of trees: reproducibility archive*, version 1.1.0, 3 September 2026. <https://doi.org/10.5281/zenodo.22263331>.
- [3] E. X. Fang, J. Lu, E. Nevo, Y. Yao, H. Zheng, *Unimodality of Independence Polynomials for Sufficiently Large Forests*, arXiv:2609.20961v1, 17 September 2026. <https://arxiv.org/abs/2609.20961>.